Aeris HR Assurance

Complete transaction coverage. Evidence-led explanations. Human decisions.

1 / Product and detection approach

Aeris replaces sampled review with repeatable checks over every supplied transaction. The example batch contains 10,000 transactions, 2,000 employees and 16 illustrative policies across hires, pay changes, allowances, transfers, leave and exits. These are demonstration policies, not verified Emirates policies.

Each active rule is considered for every record. Applicable checks produce pass, policy exception or missing/invalid evidence; irrelevant checks are not applicable. Exact comparisons, document metadata, date ordering, timing windows, grade-based thresholds and duplicate detection are deterministic. Missing evidence never silently becomes a pass. Historical employee attributes and calendar-day semantics require policy-owner confirmation.

2 / Policy intake and explanations

Written policies become inactive draft conditions through a controlled English parser. Transaction and employee CSVs, written policy Markdown and a finding-format CSV have intake paths. Supported English becomes inspectable logic; unsupported wording requires clarification. The analyst previews the impact, checks the wording and explicitly approves activation. No arbitrary model-generated code is executed.

Findings retain the transaction ID, policy ID/version, exact policy wording, field evidence, severity, recommended owner and corrective action. The interface is English-only, as requested. The original policy wording remains available for traceability.

3 / Operational and employee workflow

Analysts use severity and due-date priority, with team and named-owner filters. Operations submits fix evidence; compliance verifies closure or reopens the case. Reviewers record employee clarification or appeal context; the prototype does not contact employees. Neutral wording describes evidence rather than blaming individuals. No employee transaction is changed automatically.

Peer-pattern investigation leads are separate from policy exceptions. Their detector version, comparison group and data evidence are visible. An unusual pattern is a request to investigate, not a finding of misconduct, and is excluded from default policy-violation submission exports.

4 / Review and audit readiness

Retained batches, rule-run snapshots, review histories and workspace backup/restore support repeat review. Monthly trends combine process, team and policy filters. Policy-specific rates use applicable transactions; missing evidence is separate. Hotspots use relevant volumes. Synthetic replay reuses existing batches without duplicate findings and pauses when the browser closes.

Responsible operation & pilot

A prototype with explicit deployment and validation boundaries.

5 / Responsible AI and data handling

The prototype connects to no generative model or external AI service. The policy parser is bounded and deterministic; unsupported language requires human interpretation. Statistical leads do not decide employee outcomes. Human review is mandatory for exception disposition and policy activation.

Only synthetic data is permitted; import confirmation is a user assertion, not an automatic privacy guarantee. Supplied CSVs, policies and derived evidence stay in an IndexedDB vault on the reviewer's device; importing them pauses hosted synchronization. Local workspace backups support recovery and must remain in approved storage. Browser storage can be cleared and is not an immutable or centrally managed audit archive. Keep exports in an approved internal repository.

The latest repository includes optional private-server synchronization for generated demo-only workspaces; hosted rollout of this release is pending. Server writes require the platform's signed-in identity, reject cross-origin mutations and protected/imported batches, and use optimistic concurrency. Hosted demo audit events receive server actor identity and linked SHA-256 hashes. This is application-level tamper evidence, not a regulatory certification or independently anchored audit ledger. Persona selection demonstrates workflow; it is not enterprise role authentication.

6 / Running inside Emirates

Deploy the static interface and checking engine within the approved network. Replace demo storage with internally managed durable records and immutable audit retention. Integrate enterprise SSO and server-enforced permissions for analyst, operations owner and auditor. Add an internal scheduled ingestion worker, effective-dated policies, employee history snapshots, controlled document references and monitored backups. No live-system integration is required for this competition prototype.

If AI-assisted interpretation is added later, use only an approved private deployment with documented retention, access controls and policy-owner approval. Keep final detection logic inspectable and approved. Do not send the challenge pack to a model or retaining third-party service.

7 / Pilot plan and acceptance

Weeks 1-2: confirm schema, policy wording, severity and owner mappings; resolve ambiguities with HR compliance.
Weeks 3-4: validate planted and boundary cases, assess false positives, test permissions, privacy and restoration.
Weeks 5-6: shadow-run approved synthetic/historical scenarios, train reviewers and agree operational ownership before any production decision support.

Acceptance measures: coverage of all applicable checks; precision and recall against labelled synthetic ground truth; proportion of findings with complete evidence; assignment correctness; review/closure time; and successful replay and backup restoration. No hidden-test accuracy or pilot commitment is claimed without the official files and an agreed team.

8 / Submission readiness and remaining inputs

The runnable repository includes setup and 36 passing automated tests. The hosted link is an earlier release. The recording script targets 2:50; the earlier 72-second demo video predates the final changes. Template export preserves confirmed column mappings and excludes worked examples and investigation leads. Official transactions, employees, policies and the finding template are pending. Their schema, severity values and finding granularity need validation. Labels are needed to measure accuracy, not to generate findings. The 603 demo exceptions and 145 evidence gaps are not an official submission.